



BUDGE BUDGE INSTITUTE OF TECHNOLOGY
Affiliated to MAKAUT(formerly WBUT)& Approved by AICTE, NAAC Accredited
Kolkata - 700 137, West Bengal, India

Department of Computer Science and Engineering

June , 2026

Major Project Presentation

Topic:- Deep Learning Based Encoder and Decoder

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OBJECTIVE

- ② Develop an automated retinal blood vessel segmentation system based on the **Encoder–Decoder concept**.
- ② Use retinal fundus images from the **DRIVE dataset** for analysis.
- ② Improve image quality through preprocessing techniques such as Gaussian Blur and Thresholding.
- ② Extract retinal blood vessels accurately using the implemented segmentation workflow.
- ② Compare segmented images with expert-annotated ground truth masks.
- ② Evaluate performance using **Accuracy, Precision, Recall, Specificity, and F1-Score**.



ABSTRACT

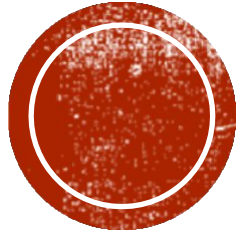
- ⑦ Develops an automated retinal blood vessel segmentation system using retinal fundus images from the **DRIVE dataset**.
- ⑦ Applies image preprocessing techniques such as **Gaussian Blur** and **Thresholding** to improve vessel extraction.
- ⑦ Segments retinal blood vessels and compares the results with expert-annotated **ground truth masks**.
- ⑦ Evaluates system performance using **Accuracy, Precision, Recall, Specificity, and F1-Score**.
- ⑦ Provides a simple and reproducible workflow that can be extended for future deep learning–based medical image analysis.



LITERATURE REVIEW

- Previous studies have shown that **retinal blood vessel segmentation** is an important step for the early detection of **DR** and other retinal diseases.
- Traditional image processing methods such as **filtering, thresholding**, are simple but often fail to detect thin blood vessels accurately.
- Recent research has focused on **Encoder–Decoder architectures**, which improve segmentation accuracy by learning complex vessel features automatically.
- Performance of segmentation methods is commonly evaluated using **Accuracy, Precision, Recall, Specificity, and F1-Score**.





Programming Language

- Python

Development Environment

- Google Colaboratory

Dataset

- DRIVE Dataset

Libraries

- OpenCV
- NumPy
- Matplotlib

- Scikit-learn

Image Processing Techniques

- Gaussian Blur
- Thresholding
- Pixel-wise Comparison

⊙ Evaluation Metrics

- Accuracy
- Precision
- Recall
- Specificity
- F1-Score

USED TECH STACK

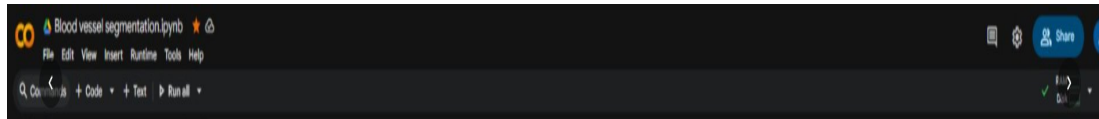
PROPOSED APPROACH

- Load retinal fundus images and corresponding ground truth masks from the **DRIVE dataset**.
- Perform **image preprocessing** to reduce noise and enhance retinal image quality.
- Apply **Gaussian Blur** and **Thresholding** to accurately extract retinal blood vessels.
- Compare the segmented blood vessels with expert-annotated **ground truth masks** using pixel-wise comparison.
- Evaluate the segmentation performance using **Accuracy, Precision, Recall, Specificity, and F1-Score**.
- Provide a simple, reliable, and reproducible workflow for automated retinal blood vessel segmentation.



Dataset & Image Loading

Development Environment



Dataset Loading

```
[4] ✓ 26s from google.colab import drive
drive.mount('/content/drive')

... Mounted at /content/drive

[5] ✓ 24s import zipfile
import os

with zipfile.ZipFile("/content/drive/MyDrive/DRIVE.zip", 'r') as zip_ref:
    zip_ref.extractall("/content/DRIVE")

with zipfile.ZipFile("/content/drive/MyDrive/A. Segmentation (1).zip", 'r') as zip_ref:
    zip_ref.extractall("/content/SEG")

print("Extraction done")

... Extraction done
```



Image Processing & Segmentation

Image Preprocessing

```
141 import os
142 import cv2
143 import numpy as np
144 import matplotlib.pyplot as plt
145
146 imatrix = []
147
148 folder = [
149     "/content/DRIVE/DRIVE/test/images",
150     "/content/DRIVE/DRIVE/training/images"
151 ]
152 for folder in folders:
153     for root, dirs, files in os.walk(folder):
154         for file in sorted(files):
155             if file.lower().endswith(('.png', '.jpg', '.tif', '.tiff')):
156                 img_path = os.path.join(root, file)
157                 img = cv2.imread(img_path)
158                 if img is None:
159                     continue
160                 green = img[:, :, 1]
161                 equ = cv2.equalizeHist(green)
162                 imatrix.append(equ)
163
164 print("Images loaded:", len(imatrix))
165
166 def createGaussianFilterBank():
167     filters = []
168     ksize = 31
169     for theta in np.arange(0, np.pi, np.pi/12):
170         kernel = cv2.getGaborKernel((ksize, ksize), 4, theta, 10, 0.5, 0, ktype=cv2.CV_32F)
171         kernel = kernel / (np.sum(kernel) + 1e-6)
172         filters.append(kernel)
173     return filters
174
175 def applyFilters(img, filters):
176     responses = []
177     for k in filters:
178         resp = cv2.filter2D(img, cv2.CV_32F, k)
179         responses.append(resp)
180     return np.max(responses, axis=0)
181
182 bank_gf = createGaussianFilterBank()
183
184 im_gauss = []
185 for equ in imatrix:
186     equ2 = applyFilters(equ, bank_gf)
187     equ2 = cv2.normalize(equ2, None, 0, 255, cv2.NORM_MINMAX)
188     equ2 = equ2.astype(np.uint8)
189     equ2 = cv2.threshold(equ2, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
190     im_gauss.append(equ2)
191
192 print("Processed:", len(im_gauss))
193
194 --- Images loaded: 40
195 Processed: 40
```

Final Blood Vessel Segmentation

```
421 import cv2
422 import matplotlib.pyplot as plt
423
424 binary_vessels = []
425
426 for img in final_vessels:
427     ret, th = cv2.threshold(img, 12, 255, cv2.THRESH_BINARY)
428     binary_vessels.append(th)
429
430 print("Processed:", len(binary_vessels))
431
432 if len(binary_vessels) > 0:
433     index = 37 if len(binary_vessels) > 19 else 0
434
435     plt.imshow(binary_vessels[index], cmap='gray')
436     plt.title("Binary Vessel Output")
437     plt.axis('off')
438     plt.show()
439 else:
440     print("No output found")
```

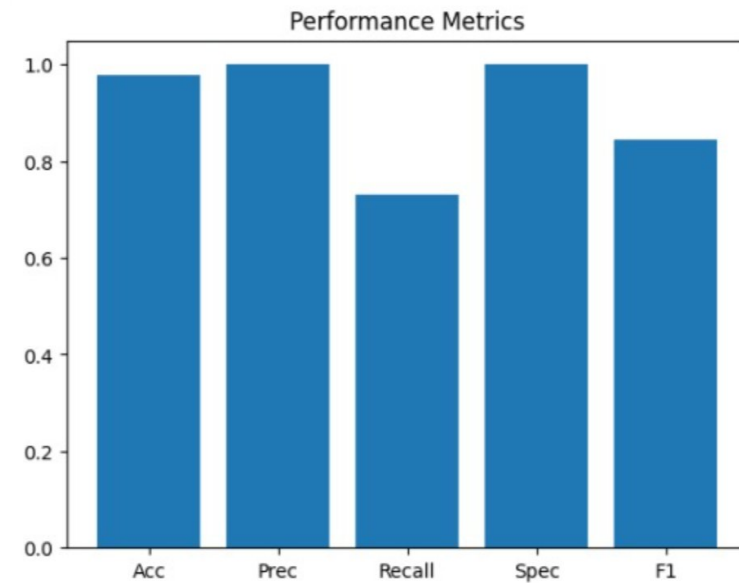


Results & Performance Evaluation

Performance Evaluation Output

```
Accuracy: 0.9786580191538369  
Precision: 0.9999999999476494  
Recall: 0.7306456548068143  
Specificity 0.9999999999967085  
F1 0.8443614446210862
```

Statistical Graph



CONCLUSION

- ⊙ Successfully developed an automated retinal blood vessel segmentation workflow using the **DRIVE dataset** and image preprocessing techniques.
- ⊙ The proposed approach effectively extracted retinal blood vessels and evaluated performance using **Accuracy, Precision, Recall, Specificity, and F1-Score**.
- ⊙ The **Encoder–Decoder concept** provides a strong theoretical foundation for advanced retinal image segmentation and future deep learning applications.
- ⊙ The proposed system can support early detection of **Diabetic Retinopathy (DR)** and serve as a foundation for future computer-aided diagnosis systems.



Reference

- ⊙ **Li, R., et al.**, "Rethinking the U-Net Architecture Design for Retinal Blood Vessel Segmentation," *IEEE Transactions on Medical Imaging*, vol. 42, no. 8, pp. 2215–2228, 2023.
- ⊙ **Zhang, Y., et al.**, "Deep Learning-Based Retinal Vessel Segmentation: A Survey," *Computer Methods and Programs in Biomedicine*, vol. 235, 2023.
- ⊙ **Staal, J., Abramoff, M. D., Niemeijer, M., Viergever, M. A., & van Ginneken, B.**, "Ridge-Based Vessel Segmentation in Color Images of the Retina," *IEEE Transactions on Medical Imaging*, vol. 23, no. 4, pp. 501–509, 2004.
- ⊙ **DRIVE Dataset (Digital Retinal Images for Vessel Extraction)**





THANK YOU